

Mathematical Foundations for Computing

Module 1

Module 1: Sets and Set Theory

Lesson 1.0 Introduction

- ✔ Course structure and navigation
Reading • 15 min
- ✔ How to learn effectively on this course
Reading • 15 min
- ✔ Introduction to the specialisation and this course
Video • 4 min
- ✔ Course Syllabus
Reading • 10 min

Lesson 1.1 Introduction to set theory

-  The definition of a set
Video • 8 min
-  The definition of a set
Reading • 10 min
-  The definition of a set
Practice Assignment • Grade: 100%
-  The listing method and rule of inclusion
Video • 8 min
-  The listing method and rule of inclusion
Practice Assignment • Grade: 100%
-  The powerset of a set
Video • 10 min
-  The powerset of a set
Practice Assignment • Grade: 100%
-  Set operations
Video • 10 min
-  Set operations
Practice Assignment • Grade: 100%

Lesson 1.2 Set representation and manipulation

-  The representation of a set using Venn diagrams
Video • 4 min
-  The representation of a set using Venn diagrams
Practice Assignment • Grade: 100%
-  Mastering Set Operations and Venn Diagram Reasoning
Dialogue • 30 min
-  De Morgan's laws
Video • 8 min
-  De Morgan's laws
Practice Assignment • Grade: 100%
-  Laws of sets: Commutative, associative and distributives
Video • 11 min
-  Laws of sets: Commutative, associative and distributives
Reading • 15 min
-  Laws of sets: Commutative, associative and distributive
Practice Assignment • Grade: 100%
-  Partition of a set
Video • 3 min
-  Partition of a set
Practice Assignment • Grade: 100%

Lesson 1.3 Summary and Assessment

- ✓ **Module 1 summary**
Reading • 15 min
 - Check your understanding: End of module 1
Graded Assignment • 20 min

Module 2

Module 2: Number Systems and Bases

Module 3

Module 3: Functions and Their Properties

Module 4

Module 4: Relations

Module 1 summary



0:04 / 2:30



Summary of set theory

Set theory is a fundamental area of mathematics that studies collections of objects, known as sets. It forms the basis for various mathematical disciplines and has applications in computer science, logic and more. Here's a concise summary of the key concepts and laws in set theory.

1. Basic concepts
 1. Definition of a set and how sets are represented. This includes the listing method, the set builder notation method and Venn diagram.
 2. Member of a set, cardinality of a set and subset of set.
 3. Types of sets: empty set, finite and infinite sets.
 4. Powerset of a set.
2. Set operations
 1. Union, intersection, set different, and symmetric difference of sets.
 2. The complement of a set.
3. Laws of set theory
 1. Commutative laws.
 2. Associative laws.
 3. Distributive laws.
 4. Identity laws.
 5. Complement laws.
 6. Absorption laws.
 7. Set difference law.
 8. De Morgan's laws.

You also need to know how use the laws of set theory or Venn diagrams to prove a relation between two given sets.

As you progress through your course, it is crucial to regularly assess your understanding and capabilities against the learning outcomes. This exercise is designed to help you reflect on your learning journey, identify areas where you may need to deepen your knowledge or skills, and develop a plan for improvement. Please follow the steps below:

1. **Review the learning outcomes.** For each of this topic's learning outcomes, rate your understanding on a scale of 1–5 (1 = very low, 5 = very high). Provide a brief explanation for your rating.
2. **Identify areas for improvement.** Review your ratings and explanations:
 1. Identify any areas where you rated your understanding below a 4.
 2. For each area identified, describe specific aspects you find challenging or unclear.
3. **Create an action plan.** For each identified knowledge gap, outline specific actions you will take to improve your understanding. Consider the following strategies:
 1. Reviewing the course materials and textbooks.
 2. Seek out additional resources that could aid your understanding.
 3. Practicing additional problems or exercises.
 4. Seeking help from the instructor or peers.
 5. Scheduling additional study sessions on the topic.

Implement your action plan and regularly review your progress. Adjust your strategies as needed based on your ongoing self-assessment and any new feedback you receive. Remember, learning is a continuous process, and it's okay to revise your plan to better suit your needs.

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